

Does Competition Undermine Higher Education Quality? Evidence from Brazil

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Abstract

Educators and regulators often argue that competition among higher-education providers, while expanding enrollment and lowering fees, comes at the cost of student achievement and programme quality. We test this claim using administrative data on Brazil's private higher education sector, defining local markets as subject-city pairs and measuring competition with a Herfindahl-Hirschman Index instrumented by its value eight years earlier. Naive estimates appear to confirm the conventional wisdom, but once we control for the ability composition of students and for programme quality, instrumented by academic staff qualifications, the direct effect of competition on achievement becomes indistinguishable from zero. The apparent achievement penalty is a selection effect — driven by the disproportionate entry of lower-ability students into an expanding market — not a quality effect. Examining the full distribution rather than the mean, we show that competition raises achievement and quality at the top while expanding access at the bottom, evidence of vertical differentiation rather than uniform decline. A market-level cost-benefit exercise shows competition raises the social benefits of higher education by more than it raises costs. Concerns about a competition-driven race to the bottom, on the margins examined here, appear misplaced.

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1 Introduction

As national higher-education systems have shifted from elite to mass provision, regulators have increasingly relied on provider competition to expand enrollment and contain tuition fees. This has fed a recurring concern among educators and policymakers: that competition, while effective on price and quantity margins, comes at the cost of student achievement and programme quality (Carvalho, 2013; Shumar, 2013). The concern has shaped real policy — including, in Brazil, a five-year moratorium on new medical programmes intended to curb the perceived effects of unchecked competition on quality.

The empirical literature on this question has been hampered by two persistent problems: omitted local and subject-specific factors correlated with both market structure and outcomes, and simultaneity between market structure and the outcomes it is meant to explain, since providers’ performance and entry decisions are jointly determined (Evans et al., 1993). A parallel debate has played out in the school-choice literature on K-12 education, where the evidence on whether competition among schools raises or lowers pupil performance is similarly mixed (Smith and Meier, 1995; Wrinkle et al., 1999); our results below suggest one reason that debate has been hard to settle.

This paper makes three contributions. First, we address both empirical problems directly: we define local markets narrowly as subject-city pairs — exploiting the horizontal differentiation of higher education programmes, an option unavailable to researchers working with school-level or district-level data such as Borland and Howsen (1992, 1993, 1996) or Hoxby (1997) — which lets us include two-way city and subject fixed effects without collinearity with our market-structure measure, and we address simultaneity by instrumenting each market’s contemporary Herfindahl-Hirschman Index (HHI) with its value eight years earlier, before any student in our sample had entered higher education. Second, we decompose the mean effect of competition on achievement into a selection channel and a quality channel, something the cross-sectional designs used elsewhere in this literature cannot do without both a pre-entry ability measure and an instrumented quality proxy. Third, we move beyond mean effects to the full distribution of both student ability and programme quality, which lets us characterize competition’s effect as vertical differentiation rather than

uniform decline or improvement — a distinction that matters directly for how the result should be read against Hoxby (1997), who finds that U.S. college-market competition raises mean enrolled-student ability and mean institutional quality. We find the opposite pattern at the mean but a similar pattern of increased differentiation, a contrast we return to in Section 5.

Naive estimates replicate the conventional concern: markets with more competition (lower HHI) show lower fees, higher enrollment, and lower mean student achievement and programme quality. But we show that this masks two distinct channels. First, a composition effect: because competition expands access primarily by lowering fees, it disproportionately draws in students who would not otherwise have entered higher education, most of whom have lower prior ability as measured by their national secondary-school examination scores. Second, a quality effect operating alongside a differentiation effect: mean programme quality falls with competition, but the full distribution of quality widens, with top programmes becoming higher quality and bottom programmes lower. Once we control for both channels — individual prior ability and lagged, instrumented programme quality — the direct effect of competition on student achievement is statistically indistinguishable from zero.

Examining the distribution rather than the mean therefore overturns the policy conclusion suggested by mean effects alone. Students at the top of the ability distribution gain access to higher-quality programmes at lower fees in competitive markets; students at the bottom gain access to higher education at all, at lower fees, even though the programmes available to them are, on average, of lower quality than the sole local option in a monopoly market. A market-level cost-benefit calculation, aggregating quality-weighted enrollment against total fees paid, shows that the social benefits of this expansion outpace its costs: moving from a monopoly market structure to near-perfect competition raises estimated social benefits by an estimated 333% against a 310% rise in costs.

The remainder of the paper proceeds as follows. Section 2 describes the institutional setting, data, and construction of the competition measure. Section 3 sets out the identification strategy. Section 4 presents results on achievement, ability composition, programme differentiation, the social cost-benefit calculation, and a battery of robustness checks. Section 5 discusses implications and concludes.

2 Institutional Setting and Data

Brazilian higher education offers an unusually clean setting for this question. Admission to most programmes is governed by a national secondary-school examination, providing a pre-entry ability measure for nearly all entrants, and a national quality-assessment exercise tests students again in their final year, providing a comparable post-entry achievement measure. With 2,152 providers and 24,955 programmes spread across 502 cities in a single national market, and with overall participation still relatively low by OECD standards, Brazil offers substantial cross-sectional variation in local market structure through which to identify the effects of competition on both participation and achievement.

2.1 Achievement and prior ability

We measure student achievement using scores on the National Examination of Student Development (ENADE), administered to students in their final year as part of the triennial national quality-assessment cycle. Because a full cross-section of subjects is only completed once every three years, we restrict the analysis to the single complete 2015–2017 cycle, and standardize scores within subject and year so that achievement is measured relative to the national distribution of students taking the same examination in the same year. We measure prior ability using each student’s score on the National High School Examination (ENEM), the standard secondary-school leaving and university-admission examination, which we take as a summary statistic for prior formal and informal education, family background, and innate ability at the point of entry into higher education. The matched sample of pre-entry and final-year scores comprises 324,616 students. We exclude public universities, which do not compete for students on price and operate at capacity irrespective of local market structure, and online programmes, whose national catchment area does not fit a local market definition.

2.2 Programme quality

Direct measures of programme quality — such as staff-student ratios or salaries — are poor proxies for the unobservable components of quality that plausibly matter most for

achievement (Goldhaber and Brewer, 1997; Vignoles et al., 2000). We instead use the average final-year score of the previous cohort in the same programme as an omnibus quality proxy, capturing both observable and unobservable quality. Because this measure could itself be confounded by the same selection effects we are trying to net out, we instrument it with the share of academic staff in the programme holding a postgraduate qualification.

2.3 Measuring competition: markets and the HHI

We define a local market as a subject-city pair, treating two providers as competitors only if they offer the same subject in the same municipality. This departs from the standard practice of measuring concentration at the geographic level alone (as in Borland and Howsen, 1992 and Hoxby, 1997) and is important for identification: because not all subjects within a city are substitutes, defining markets at the subject-city level lets us include both city and subject fixed effects simultaneously, without the collinearity between market-structure measures and geographic fixed effects that constrained earlier studies using geography-only market definitions.

Restricting markets to the municipal level is defensible on institutional grounds. Most classroom-based students in our data live with parents (61%) or partners (26%), attend class in the evening (80%) while working during the day (64%), and cite proximity to home or work as their primary reason for choosing an institution (24%, per the 2017 ENADE student questionnaire) — all consistent with limited willingness to commute across municipal boundaries for a first-choice programme. We compute the Herfindahl-Hirschman Index of enrollment shares within each subject-city market using the OECD 6-digit subject classification to define comparable course groupings. Table 1 summarizes the resulting data sources and sample sizes, which vary across specifications depending on data availability for each variable.

3 Identification Strategy

Standard microeconomic theory predicts that, relative to a monopoly, competitive equilibrium features lower prices and higher quantities; quality can move in either direction

Table 1: Data sources

Measure	Source	Coverage
Final-year achievement (ENADE)	INEP quality-assessment micro-data	2015–2017 cycle, 324,616 students
Prior ability (ENEM)	INEP national exam microdata, matched to ENADE	Same sample, matched by student
Programme quality proxy	Lagged cohort ENADE mean; academic staff qualification (instrument)	Previous (2012–2014) assessment cycle
Market structure (HHI)	Higher Education Census enrollment shares by subject-city market	Contemporary and 2009 (instrument)
Fees and enrollment	Higher Education Census	Subject-city market level, 4,988–8,469 markets depending on specification

depending on the model. Because market structure and outcomes are jointly determined — provider performance feeds back into entry and market shares — a naive regression of outcomes on contemporary HHI is subject to simultaneity bias (Evans et al., 1993). We address this with an instrumental-variables strategy: we instrument each market’s contemporary HHI with its value in 2009, the year before any student in our sample entered higher education. Because our data forms a single cross-section rather than a time series, the usual concern that lagged instruments inherit the autocorrelation of the outcome does not apply here (Wang and Bellemare, 2019).¹

We first validate the instrument and the HHI measure itself by checking that it predicts market outcomes in the direction implied by theory:

$$\text{Fee}_{sc} = \alpha + \beta \text{HHI}_{sc} + \sum_c \theta_c \text{City}_c + \sum_s \gamma_s \text{Subject}_s + \varepsilon_{sc} \quad (1)$$

¹For the same reason, the recent critique of two-way fixed-effects estimators in settings with staggered treatment timing (Imai and Kim, 2021; Wooldridge, 2021) does not apply to our design: our city and subject fixed effects partition a single cross-section of markets rather than a panel with variation in treatment timing across periods.

$$\text{Enrollment}_{sc} = \alpha + \beta \text{HHI}_{sc} + \sum_c \theta_c \text{City}_c + \sum_s \gamma_s \text{Subject}_s + \varepsilon_{sc} \quad (2)$$

Table 2: Market-level validation: effect of concentration on fees and enrollment

	Fees			Enrollment		
	OLS	IV	1st stage	OLS	IV	1st stage
HHI	61.9**	298.5***	—	−252.0***	−1378.3***	—
HHI (2009, instrument)	—	—	0.42***	—	—	0.42***
Subject / city FE	Yes/Yes	Yes/Yes	No/No	Yes/Yes	Yes/Yes	No/No
Observations	4,988	4,988	4,988	4,988	4,988	4,988
1st-stage F -statistic	—	—	174.65	—	—	174.65

Note: standard errors clustered at municipal and subject level (in online appendix). ** $p < 0.05$, *** $p < 0.010$. The first-stage F -statistic of 174.65 comfortably exceeds the Staiger and Stock (1997) threshold of 10, while remaining well below levels that would suggest the eight-year lag is simply a linear proxy for the contemporary HHI.

The results confirm the instrument’s validity and the HHI’s meaningfulness as a competition measure: more concentrated (less competitive) markets show significantly higher fees and lower enrollment, exactly as monopoly theory predicts. If anything, failing to correct for simultaneity would bias estimates conservatively, since unobserved competitive pressure is partly absorbed by the subject and city fixed effects already in the model.

4 Results

4.1 Competition and student achievement

Table 3 estimates the effect of HHI on final-year achievement, adding controls for prior ability and instrumented programme quality successively.

Column IV(1), with no controls, replicates the conventional concern: more competition (lower HHI) is associated with lower achievement. Adding the individual prior-ability control in IV(2) sharply attenuates the coefficient, and adding the instrumented programme-quality control in IV(3) drives it to a value statistically indistinguishable from zero. Once both channels through which market structure could plausibly affect a student’s final score are accounted for, competition itself has no residual direct effect on achievement. This pattern is,

Table 3: Effect of market competition on student achievement

	OLS (1)	OLS (2)	OLS (3)	IV (1)	IV (2)	IV (3)
HHI	0.081**	0.065**	0.045	0.34***	0.20***	0.001
Prior ability (ENEM)	—	0.58***	0.57***	—	0.58***	0.56***
Programme quality (IV)	—	—	0.091***	—	—	0.19***
Subject / city FE	Yes/Yes	Yes/Yes	Yes/Yes	Yes/Yes	Yes/Yes	Yes/Yes
Observations	324,616	324,616	324,616	324,616	324,616	324,616

Note: standard errors clustered at market level (in online appendix). ** $p < 0.05$, *** $p < 0.010$.

if anything, more supportive of a well-specified model than the mixed and inconclusive results that have characterized this literature (Belfield and Levin, 2002); the two-way subject-city fixed effects available in our market definition were not feasible in the school-district-level data used by Borland and Howsen (1992, 1993), whose model may for that reason have left local heterogeneity unaddressed in their market-structure coefficient.

4.2 Composition of student ability

Table 4 regresses individual prior-ability scores directly on HHI, with city and subject fixed effects, both for the full sample and separately for the top and bottom decile of entry scores within each market.

Table 4: Effect of competition on student ability composition

	OLS			IV		
	All	Top decile	Bottom decile	All	Top decile	Bottom decile
HHI	0.029	−0.48***	0.61***	0.25**	−1.16***	1.60***
Observations	324,616	47,643	47,643	324,616	47,643	47,643

Note: standard errors clustered at municipal and subject level (in online appendix). ** $p < 0.05$, *** $p < 0.010$.

The mean coefficient masks offsetting movements at the tails. In more competitive markets, the bottom decile of entrants has lower prior ability than the bottom decile in less competitive markets — consistent with an expansion of access to marginal students — while the top decile of entrants has higher prior ability, consistent with better-matched sorting of

high-ability students into more differentiated competitive markets. Both tails move toward students being better off: lower-ability students gain access to higher education they would otherwise lack, and higher-ability students are not displaced. Re-estimating the achievement model of Table 3 separately by ability decile confirms the pattern: higher-ability students achieve higher final scores in more competitive markets, while lower-ability students' scores fall, but for the latter group the relevant counterfactual is likely non-participation rather than a higher score in a monopoly market.

4.3 Vertical differentiation of programme quality

Table 5 repeats this exercise using our instrumented programme-quality measure as the dependent variable, again splitting the sample into the top- and bottom-quality programme within each market.

Table 5: Effects of competition on programme quality, by market position

	OLS			IV		
	All	Top quality	Bottom quality	All	Top quality	Bottom quality
HHI	0.12	−0.70***	0.84***	1.26***	−1.16***	2.95***
Observations	8,469	4,638	4,602	8,469	4,638	4,602

Note: standard errors clustered at municipal and subject level (in online appendix). *** $p < 0.010$.

As with student ability, mean programme quality falls with competition, but the effect is not uniform: the top-quality programme in a competitive market is of higher quality than its counterpart in a less competitive market, while the bottom-quality programme is of correspondingly lower quality. This is vertical differentiation, not uniform decline. Table 6 shows that fees fall with competition for both top- and bottom-quality programmes, meaning that higher-ability students gain access to better-matched, higher-quality programmes at lower fees in competitive markets, while lower-ability students gain access to some programme, also at lower fees, where previously they may have had none.

Table 6: Fee differentiation by market position

	OLS			IV		
	All	Top quality	Bottom quality	All	Top quality	Bottom quality
HHI	47.2*	73.9**	78.7***	311.4***	345.1***	350.8***
Observations	8,469	4,638	4,602	8,469	4,638	4,602

Note: standard errors clustered at municipal and subject level (in online appendix). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.010$.

4.4 A social cost-benefit calculation

The results so far establish that competition redistributes access and achievement across the ability distribution rather than uniformly eroding quality. To assess whether this redistribution is, on net, socially desirable, we follow Hanushek and Woessmann (2015) and UNESCO’s Education at a Glance (UNESCO, 2018) in constructing market-level measures combining enrollment and quality. Total benefit is the log of mean final-year scores in a market weighted by freshman enrollment, summed across programmes in the same subject-city market; total cost is the log of aggregate tuition fees paid, similarly summed.

Table 7: Effect of competition on total social benefits and costs

	OLS: Benefit	OLS: Cost	IV: Benefit	IV: Cost
HHI	−1.29***	−1.21***	−3.33***	−3.10***
Observations	4,988	4,988	4,988	4,988

Note: standard errors clustered at municipal and subject level (in online appendix). *** $p < 0.010$.

The IV coefficients imply that moving from monopoly ($\text{HHI} = 1$) to near-perfect competition ($\text{HHI} \approx 0$) raises total social benefits by approximately 333% against a 310% rise in total costs — that is, competitive markets deliver more social benefit from higher education per unit of cost than concentrated ones. Combined with the distributional results above, this suggests that the net social effect of competition in this sector is positive: more students are served, at lower cost per unit of measured benefit, without the uniform quality erosion implied by looking at mean achievement alone.

4.5 Robustness

We subject the achievement result in Table 3 to four checks, reported in full in the online appendix. First, because the distribution of HHI is skewed by a large mass of single-provider (monopoly) markets, we re-estimate Equations 1–2 and the achievement model with an added dummy for monopoly markets; the null direct effect of competition on achievement is essentially unchanged (IV coefficient of -0.015 , insignificant, against 0.001 in the baseline), indicating the result is not an artefact of how monopoly markets are treated. Second, re-estimating achievement using raw, non-standardized ENADE and ENEM scores in place of the within-subject-year standardized measures again yields a small, insignificant IV coefficient (0.27), confirming the result does not depend on the standardization. Third, we test whether the instrumented quality proxy is doing real work: replacing it with the uninstrumented lagged cohort score, or with the raw share of staff holding a postgraduate qualification, leaves a residual, statistically significant coefficient on HHI (0.091 and 0.19 respectively). The full null result is therefore specific to using a properly instrumented measure of programme quality; a noisier or endogenous quality proxy would have left an econometrician with the false impression of a residual quality penalty from competition, which is itself an informative methodological result for future work in this area.

Finally, quantile regressions of student entry scores on HHI, estimated across the full sample rather than by market-level split, show a positive and fairly stable HHI coefficient across the 1st decile, median, and 9th decile alike (IV coefficients of 0.26 , 0.25 , and 0.25 , respectively). This appears, at first glance, to sit awkwardly next to the split-sample result in Table 4, where the sign of the HHI coefficient flips between the top and bottom decile. The two exercises answer different questions: the split-sample estimates rank students *within* each subject-city market before comparing top- and bottom-ranked entrants across markets, while the pooled quantile regression ranks students across the entire national distribution regardless of market. Because competitive markets attract both higher- and lower-ability entrants than monopoly markets typically serve, a market-level ranking and a national-level ranking need not agree on where in the distribution any given competitive effect shows up. We view the two results as complementary rather than contradictory, but flag the pooled

quantile approach as a natural robustness dimension for follow-up work using a market-clustered quantile estimator.

5 Discussion and Conclusion

Educators and regulators frequently treat falling mean test scores in expanding, competitive higher-education markets as *prima facie* evidence of a race to the bottom. This paper shows that the inference does not survive a look at the underlying mechanism. Once individual prior ability and instrumented programme quality are controlled for, competition has no direct effect on student achievement; the mean decline in achievement associated with competition is a composition effect, driven by the entry of students who would not otherwise have participated in higher education at all. Examining the distribution of both student ability and programme quality shows that competition drives vertical differentiation rather than uniform decline: high-ability students are matched to higher-quality, lower-fee programmes, while low-ability students gain access to some higher education, also at lower fees, where a monopoly market might have offered none. A market-level cost-benefit calculation indicates that the resulting expansion in access delivers social benefits that outpace its costs.

These results diverge in part from Hoxby (1997), who finds that U.S. college-market competition raises the mean ability of enrolled students and raises mean institutional quality; we find the opposite pattern at the mean, though a similar pattern of increased vertical differentiation. A plausible reconciliation is that the Brazilian setting we study features much lower baseline participation rates than the U.S. context Hoxby examines, so that competition’s dominant effect operates through the extensive margin of participation rather than the selective re-sorting of an already-participating population. This suggests that the effects of competition on higher-education outcomes may depend systematically on a system’s stage of expansion — a hypothesis this paper’s cross-sectional design cannot test directly, but which offers a natural direction for future comparative work.

Two limitations bound the interpretation of these results. First, our achievement and quality data are drawn from a single completed three-year assessment cycle, so the analysis is cross-sectional; we cannot rule out longer-run dynamic effects of sustained competitive

pressure on quality that a panel spanning multiple cycles might reveal as data accumulates. Second, our achievement measure, like all standardized test scores, is only a partial proxy for the value higher education adds (Hanushek, 1979, 1986); it says nothing about labour-market outcomes. A natural extension, made feasible by Brazil’s linked administrative employment records (RAIS), is to examine whether the participation gains documented here translate into measurable labour-market outcomes for the marginal students that competition brings into the system — the subject of ongoing follow-up work.

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